



(10) Publication No: **IN202641062255 A1**

(22) Date of Filing: 16-05-2026

(43) Publication Date: 29-05-2026

Journal No: 22/2026

(19) Intellectual Property Office, India

(12) INDIAN PATENT APPLICATION

(54) Title: **Hybrid ble-lora infrastructure-independent communication system for long-range iot messaging without internet or gateways**

(51) International Classification:

**H04W 84/18, H04W 4/80, H04L 12/28, H04W 40/24
H04W 4/00**

(21) Application No: **202641062255**

(31) Priority Document No: --

(32) Priority Date: --

(86) International Application No: --

Filing Date: --

(87) International Publication No: --

(61) Patent of Addition to Application No: --

Filing Date: --

(62) Divisional to Application No: --

Filing Date: --

(71) Name of Applicant(s):

**Vallurupalli Nageswara Rao Vignana Jyothi
Institute Of Engineering And Technology
(vnrvjiet),**

(72) Name of Inventor(s):

- 1. Dr. Srinivas Kanakala**
- 2. Dr. A Brahmananda Reddy**
- 3. Akula Chetan Krishna Sai**
- 4. Amaraboina Badhri**
- 5. Arisenapally Ananya**
- 6. Murala Venkata Siva Surya**

(57) Abstract:

The present invention relates to a hybrid wireless communication system enabling infrastructure-independent long-range data transmission. The system comprises a plurality of autonomous nodes, each integrating a microcontroller, a LoRa transceiver, a Bluetooth Low Energy (BLE) interface, and a power management module on a compact hardware platform. A user device communicates with a node via BLE, and the node transmits data over LoRa to other nodes in a multi-hop manner. Within each node, the message is selectively retransmitted in accordance with several criteria such as maximum hop count, distance-based filtering. The invention also provides mechanisms for duplicate filtering utilizing message-ID's, randomly timed retransmissions and channel sensing to reduce repeated message transmission and collisions. The invention can form and heal itself without outside assistance or network infrastructure, thus it is suitable for remote, off-grid, disaster throughout environments.